

Nathan Liu

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EDUCATION

University of California, Berkeley

Bachelor of Arts in Computer Science & Data Science

Berkeley, CA

May 2026

TECHNICAL SKILLS

Languages: Python, Java, SQL (PostgreSQL), C, C++, JavaScript, Go

Cloud & Systems: Docker, Kubernetes, AWS (Kinesis, Lambda, S3, EC2)

Data & ML: PyTorch, NumPy, pandas, Matplotlib, TensorFlow, LangChain

Web: React, Node.js, Django, FastAPI, MongoDB, Web3

EXPERIENCE

Data Engineer Intern

May 2025 – Aug. 2025

Carbon Sustain

Berkeley, CA

- Engineered a real-time LLM anomaly detection pipeline that processed 100k events, enforced schema-validated JSON, and logged full traces to S3 for replayability, enabling faster incident investigation.
- Enhanced system reliability by implementing DynamoDB idempotency keys to handle retry spikes and configuring CloudWatch alarms, cutting real-time execution failures by 50%.
- Cut dashboard data retrieval time by 80% by rewriting high-cost joins, adding composite indexes, and implementing cursor pagination.

Software Engineer

Jan. 2025 – Aug. 2025

UnifIBD

Remote (San Jose, CA)

- Architected a mobile CI/CD pipeline using Fastlane and GitHub Actions, automating deployments to TestFlight/Google Play to cut release times from hours to minutes.
- Accelerated time-to-production for data features by 50% by standardizing integration workflows and documentation between analytics and engineering teams.
- Led a three-person engineering team in migrating a Swift iOS app to Flutter and establishing architectural direction, technical milestones, and reduced maintenance effort.

Data Science Intern

Jun. 2024 – Aug. 2024

L.A. Lucky Import & Export Inc.

Los Angeles, CA

- Reconstructed a real-time SAP inventory sync for 4,000+ SKUs using incremental pulls and upserts into PostgreSQL which reduced manual entry by 100% and enabled automated invoicing for all customers.
- Overhauled a legacy platform (Django REST, JWT, React) and improved performance by 75% by reducing payload size and adding server-side caching.
- Built a Docker-based CI/CD workflow to deploy ETL/ELT jobs automatically, streamlining legacy database migration and improving pipeline reliability.

PROJECTS

Live Semantic Market | *JavaScript, Python, React, PostgreSQL*

- Engineered a streaming data pipeline processing and transforming global research feeds via SSE, projecting multi-dimensional vector drifts with sub-100ms latency.

Clickbait Classifier | *Python, PyTorch, Pandas*

- Fine-tuned a BERT-based classifier on 18K social-news titles; achieved 0.89 F1 and 91% accuracy on a held-out test set, outperforming a TF-IDF logistic regression baseline by +0.14 F1 points.

ItineraryAI | *Python, LangChain, FastAPI, JavaScript*

- Built a chat-based travel concierge that generates structured itineraries from a single prompt and shipped a web dashboard for trip planning (budgets, maps, calendar sync).